

Guinea Current model validation

Field	Entry
Region	Guinea Current, LME_028; Sea Around Us target boundary. Regional workbook
Catch source	Sea Around Us v 50.1 catch series, 1950–2019; retained API 88 material dated 3 September 2026. Reference: 2019 landings, including reported/unreported retained catch; discards separate. Raw catch Taxon/TL source
Selected article	Guénette & Diallo (2004), Addendum: Modèles de la côte guinéenne, 1985 et 1998, pp.124–159. In Palomares & Pauly (eds.), West African marine ecosystems: models and fisheries impacts, FCRR 12(7). Chapter PDF Paper folder
Other known articles	Gascuel, Guénette, Diallo & Sidibé (2009), Impact de la pêche sur l'écosystème marin de Guinée – Modélisation EwE 1985/2005, FCRR 17(4), 60 printed pages. The title says 1985/2005; Ecopath construction and Table 3 specify 1985/2004. 2009 report
Selected model	28_646_Guinea_(1998); 1998, waters off the Republic of Guinea, 111,932 km ² . 44 ecological source groups plus synthetic diet_import 45. This is a national coastal/offshore model transferred to the full LME. Selected JSON Model folder
Other models from the same article	The chapter also presents a 44-group 1985 counterpart derived from balanced 1998, with reconstructed early catches. Stored 28_726_Guinea_(1985) is the distinct 35-group, 42,969 km ² shelf revision in the 2009 report. Hash-suffixed copies are extraction identities, not extra baselines. Stored 1985 identity
Selection rationale	Article: ? (separate article-choice rationale unrecorded). Model: recorded rationale is geographic transfer requiring overlap validation. The source supports preferring 1998 over 1985 because contemporary landing/biomass estimates were available; the earlier model has greater catch-reconstruction uncertainty.
Model extraction	Accepted canonical parameters retained. Native database JSON differs only in 44 added taxonomy descriptions. Table 9 (p.143/PDF 147) has 13 B/EE values outside strict printed-rounding bounds; PB/QB agree at displayed precision. Oiseaux TL 3.77 differs from accepted calculated 3.545. Coarse diets and reconstructed catch/discards remain limitations. Inherited diet normalization changes cells by at most 3.07×10^{-8} ; default GS, derived BA and import 45 remain accepted conventions. Source/runtime comparison Table9 differences Retained departures and evidence limits: model notes
GE diagnostics	OK. No negative SPPR entries across the basal-source columns, including unfished groups. $\rho_{\text{living}} = 0.4497$; $b = 0.3548$; Detritus (44) SPPR = 1.7792.
TE diagnostics	WARN: the deep large shark/ray pool has an extreme coefficient; saved annual TE results remain unavailable under the implausibility flags. No negative SPPR entries across the basal-source columns, including unfished groups. $\rho_{\text{living}} = 0.7176$; Detritus (44) SPPR = 0.4259.

Field	Entry
SPPR calculation	Removed groups from calculation: - Dolphins (GE=0.50%, EE=0.4, TL=4.7, sppr_TE=500K, sppr_GE=100K) - Birds (GE=0.30%, EE=0.36, TL=3.5, sppr_TE=345K, sppr_GE=66K) - Whales (GE=0.40%, EE=0.5, TL=4.0, sppr_TE=275K, sppr_GE=66K)
Geographic fit	Approximate A: 5.0–7.0% of the region covered by the study. B: 95–100% of the study inside the region.
Temporal fit	Static 1998 coefficients and historical mixture weights are applied across 1950–2019. The reference year is 21 years later; annual ecosystem/fleet changes are not reconstructed.
Other	TE retains its implausibility warning: the deep large shark/ray pool has an extreme coefficient. Symbolic negative-source failures, Monte Carlo implausibility flags and unavailable source decompositions remain. Full catch mapping does not establish whole-LME ecological validity. Method-specific exposure
Open issues and next action	[Researcher: open issues and next actions]
Review and reproducibility	Researcher name: Ido Carmel review date: 08/10/2026 MODEL VALIDATED

Taxon mapping and coverage

Reference: 2019 landings; all exact catch labels, including unidentified and zero-catch records; no taxon or group filter.

44 source ecological groups (+1 computational import); 375 catch labels.

Simple-chain PPR totals 202,194,055.86 t C in 2019.

50 labels lack a saved classic coefficient/TL; all have zero 2019 landings and therefore zero annual PPR. Unavailable TL remains ?, not zero.

Method: independent classic/simple trophic chain using the saved taxon coefficient.

[Full taxon mapping appendix](#) and [descriptive Sources sheet](#)

Overall confidence	Taxa (n)	Catch (%)	Simple-chain PPR (%)
High	159	37%	51%
Medium	107	17%	24%
Low	32	7.0%	9.3%
Very low	77	38%	16%
Unresolved	0	0%	0%

Overall confidence uses the weaker required membership or allocation component.

Group assignment rules

Plain-language rule	Confidence	PPR percentage
Exact source member in every documented applicable group.	High	49%
Complete actual source representative union supports a higher label or stage mixture; local composition inferred.	Medium	30%
Broad reporting category approximated over compatible named and residual source pools; caught composition unknown.	Very low	15%
Taxonomic extension from source congeneric representatives; unlisted membership assumed.	Medium	3.0%
Verified taxon fits the explicit source compartment without a competing specific placement.	High	1.3%
Closest positive taxonomic or habitat analogue; unrepresented habitat, size or taxon mismatch disclosed.	Very low	0.60%
Verified accepted synonym connects the catch label to a printed member.	High	0.41%

Membership evidence: [source members, taxonomy and complete candidate decisions](#) Broad bony fish includes named stocks and deep/pelagic members; dedicated cartilage groups are excluded. Elasmobranchii excludes chimaeras. Broad invertebrate sets review actual euphausiids, benthos and larval stages. Scaridae denotes the historical parrotfish family, now Scarinae within Labridae; the genus-misspelling homonym is excluded. [Scaridae rank bridge](#)

Allocation weight rules

Plain-language rule	Confidence	PPR percentage
One reviewed group; no numerical split.	High	75%
Complete accepted 1998 group-export catch proportions; historical mixture/catchability assumed.	Medium	16%
Retained whole-period single-group donor total-catch proxy; reporting/population mismatch assessed separately.	Low	9.5%

All 71 split labels retain explicit assumptions: 37 use complete 1998 model catch proportions; 34 retain historical donor total-catch weights at Low confidence. These mixtures are transferred across 1950–2019 and catch bases. [Allocation and donor applicability evidence](#)

Each rule percentage is the share of independent simple-chain PPR associated with taxa using that rule.

No unresolved taxa remain. The 77 Very low decisions cover 38.4107% of landings and 15.6210% of simple-chain PPR; the table below identifies the source-definition, composition and analogue uncertainties.

Very low decisions

Affected taxa	Why confidence is very low
Actinopterygii; Marine finfishes not identified; Marine fishes not identified	All 29 bony-fish pools include named stocks, pelagic and deep guilds; dedicated cartilage pools are excluded. Actual caught composition is unknown. Complete 1998 model catch proportions are a whole-guild proxy transferred across 1950–2019 and catch bases.
Scombroidei	Historical reporting boundaries span named and residual source fishes but differ from modern taxonomy. The actual species mixture is unknown. The source-member union and complete 1998 group catch proportions are assumptions transferred across years and bases.
Alepisaurus ferox; Aphanopus intermedius; Berycidae; Beryx; Beryx decadactylus; Beryx splendens; Brama brama; Bramidae; Gephyroberyx darwinii; Guentherus altivela; Lampris guttatus; Promethichthys prometheus; Ruvettus pretiosus	Deep-water fish are weak predatory, habitat or family analogues of the source Gempylus, Lepidocybium or other bathy members. Exact placement and size/depth boundaries remain unverified; mixed diets and pelagic overlap weaken the pooled pathway. Lampris also has an uncertain local role.
Apsilus fuscus; Brotula barbata	The snapper and brotula are weak shelf-predator analogues. Apsilus is not established by the source coastal Lutjanus members; Brotula also overlaps deeper and crustacean-feeding compartments.
Batoidea; Chondrichthyes	The four shark/ray pools lack observed species, size and habitat composition. Chondrichthyes additionally includes chimaeras absent from the source. Historical donor catch proportions mix guild members and reporting populations; they do not establish residual composition and are transferred across years/bases.
Perciformes	Historical reporting boundaries span named and residual source fishes but differ from modern taxonomy. The actual species mixture is unknown. The source-member union and complete 1998 group catch proportions are assumptions transferred across years and bases.
Aetobatus narinari; Carcharias taurus; Carcharodon carcharias; Galeorhinus galeus; Ginglymostoma cirratum; Glaucostegus cemiculus; Pristidae	Large coastal sharks/rays are weak size, habitat or family analogues of source coastal elasmobranch members. These taxa are not explicitly listed, and compartment limits are incompletely verified; white-shark regional representativeness is also unverified.
Marine pelagic fishes not identified	Source pelagic members include named tuna/billfish, clupeids and mixed shelf/bathy guilds. Their proportions within this reporting label are unknown; whole-pool 1998 catch weights transfer an assumed mixture across years/bases.
Lithognathus mormyrus	The sea bream is a weak coastal benthic-invertebrate-feeder analogue; source membership and the precise feeding/size boundary remain unverified.
Elagatis bipinnulata; Tylosurus crocodilus	These large pelagic fishes are weak feeding/habitat analogues of source Alectis/Coryphaena or Ablennes. Exact source membership and size/habitat limits are unverified.

Affected taxa	Why confidence is very low
Branchiostegus; Lagocephalus laevigatus; Parapristipoma octolineatum; Spondyllosoma cantharus	Shelf benthic feeders are weak analogues of source members such as Balistes, Plectorhinchus, Pagellus and Boops. No explicit placement is documented; the puffer overlaps pelagic/benthic guilds and source puffers span several compartments.
Sarotherodon galilaeus; Sarotherodon melanotheron	Tilapia plant/detritus feeding links weakly to source Liza/Mugil and Acanthurus/Sarpa. These are not mullets or source members; freshwater/brackish or lagoon habitats are transferred to marine pathways without a freshwater compartment.
Bathygobius soporator; Gobiidae; Solea	Gobies and soles are weak shallow benthic-feeder analogues of the source coastal guild. No explicit goby compartment exists; Solea species/depth composition is unknown. The provider large-reef bin does not establish Bathygobius placement.
Miscellaneous marine crustaceans	Actual source invertebrates include benthic taxa, euphausiids and unspecified larvae. Provider bins do not prove adult-only composition. Larval taxonomy and caught-stage composition are unobserved; source whole-pool catch weights are fixed across years/bases.
Scorpaeniformes	Source scorpionfish, gurnard and flying-gurnard members span coastal, shelf and bathy groups. Legacy order boundaries and unidentified caught composition remain uncertain; 1998 whole-group catch weights transfer the mixed pathways across years/bases.
Centrophorus; Centrophorus granulosus; Centrophorus lusitanicus; Centrophorus squamosus; Centroscyrnus coelolepis; Centroselachus crepidater; Ceterorhinus maximus; Dalatias licha; Deania calceus; Echinorhinus brucus; Hexanchus griseus; Rostroraja alba; Scymnodon ringens; Somniosus rostratus	These are weak size, depth or family analogues of source large deep elasmobranchs; exact membership and size cutoffs are unverified. Basking shark follows the named whale-shark analogue despite pelagic filter feeding; skate catch depths and some local geographic fits remain uncertain.
Centrolophidae; Peristediidae; Schedophilus ovalis; Schedophilus pamarco	Barrelfishes and armoured gurnards are weak benthopelagic-feeder analogues of source Ariomma or Trigla. Heterogeneous diets, uncertain local feeding and shelf/slope overlap leave source-specific placement unresolved.
Pellonula leonensis	Pellonula provides a weak small coastal-pelagic feeding analogue, but river/lagoon and brackish occurrence complicate transfer. Marine catch does not establish a shared freshwater food web or documented source membership.
Pseudocaranx dentex; Uraspis secunda	Trevally/jack are weak analogues of source Caranx/Trachinotus. Offshore habitat overlap and incomplete source guild boundaries prevent stronger membership confidence.
Holocentrus adscensionis	The reef squirrelfish is a weak nocturnal invertebrate-feeding analogue of source Sargocentron; exact membership and the small/medium compartment boundary are unverified.

Affected taxa	Why confidence is very low
Tetrapturus pfluegeri	The spearfish is linked to the source billfish members in Thons majeurs, but is not a printed member; source-specific placement remains a weak analogue.
Dicentrarchus; Dicentrarchus punctatus	Sea bass are weak coastal/estuarine piscivore analogues. They are not source-listed and cannot be assigned literally to Bars, which denotes Pseudotolithus; exact guild boundaries remain uncertain.
Miscellaneous aquatic invertebrates	Actual source invertebrates include benthic taxa, euphausiids and unspecified larvae. Provider bins do not prove adult-only composition. Larval taxonomy and caught-stage composition are unobserved; source whole-pool catch weights are fixed across years/bases.
Centroscyllium fabricii; Oxynotus centrina; Scyliorhinidae	Small/medium deep sharks are weak analogues of source Etmopterus/Galeus. Family depth composition and group size boundaries remain uncertain; documented Sierra Leone occurrence prevents excluding Centroscyllium merely by geography.
Chimaeriformes	Chimaeras are absent from the source inventory and belong to Holocephali rather than sharks/rays. Deep cartilaginous pools are a weak depth/body-form analogue; coastal and bony alternatives fit less closely. Source catch proportions fix an unobserved mixture across years and bases.
Cyprinidae	Freshwater carp/barbel bottom-feeding, omnivory and detritivory provide a weak link to shallow Mulets and coastal invertebrate feeders. The marine model has no freshwater compartment or measured family composition; source catch weights transfer those pathways across all years/bases.
Inermiidae	Bonnetmouth schooling and plankton feeding connect to the small pelagic guilds, but Western Atlantic identity and absence of a printed member weaken regional transfer. Model catch weights assume an unobserved coastal/offshore mixture across years and bases.
Pentheroscion mbizi	The shelf croaker has a weak source medium-predator habitat connection, supported by 80–200 m occurrence and fish/shrimp feeding. Exact membership and pooled diet fit remain uncertain.

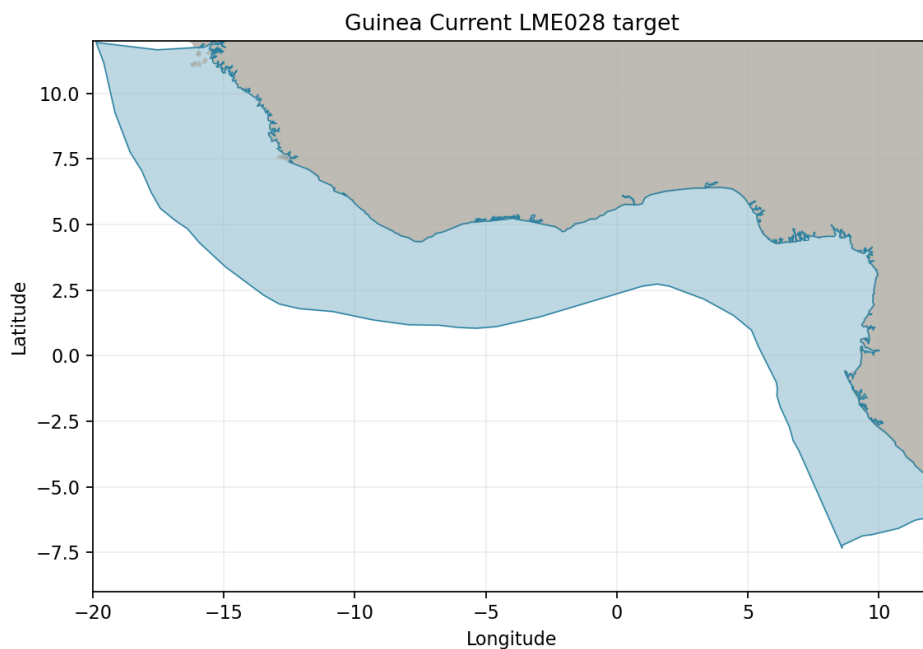
The excluded TE arithmetic is 4.5717 billion t C: Elasmobranchii contributes 23.6895%, Prionace glauca 13.2955% and Sphyrna zygaena 10.0636%. Explicit shark membership is retained; the extreme pooled coefficient and Low donor mixture make TE scientifically implausible. [Actual method contributions](#)

Geographic evidence screenshots

R is the full Guinea Current LME; S is the study waters off Guinea. The graticule trace and ordered coastline approximate the original figure. Nine source-visible water controls test both sides of the maritime limits and coastal water; these ranges are not statistical intervals.

Region area

Guinea Current target region



Sea Around Us LME 28 target R: 1,918,762 km². [Target boundary](#) | [Map image](#)

Study area in the article

Original Guinea study area Figure 1

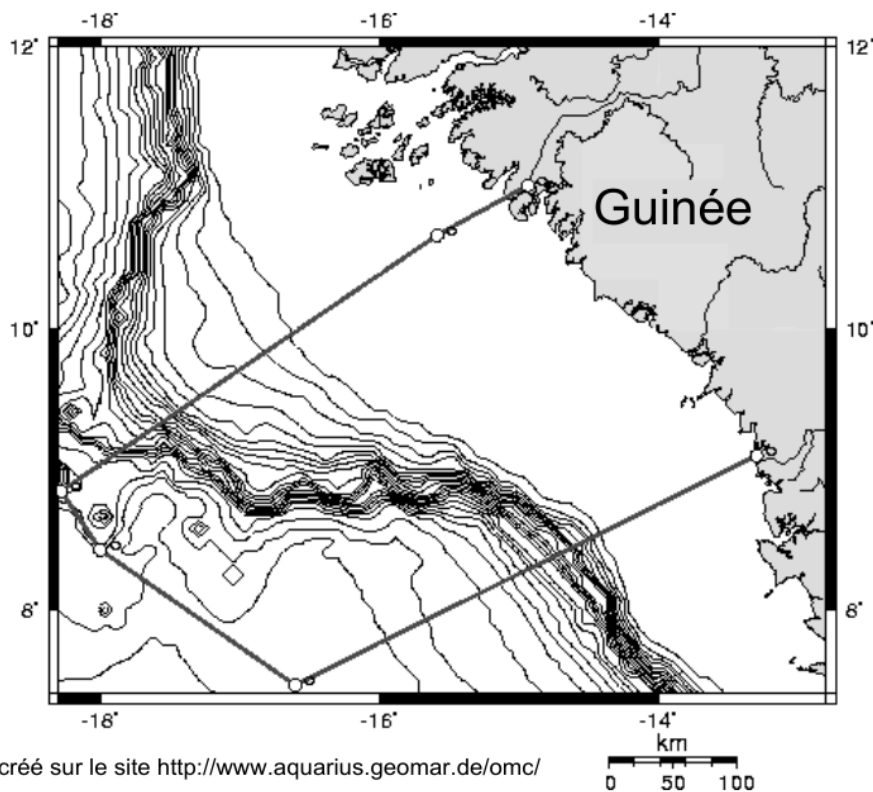


Figure 1. Localisation de la zone considérée pour le présent modèle de la Guinée.

Guénette & Diallo 2004 Figure 1, printed p.124/PDF 128. The model covers waters off the Republic of Guinea; the LME continues far beyond the study. Source-reported area is 111,932 km². [Original source](#) | [Geographic assessment](#)

Approximate source-boundary comparison: reconstructed marine area 109,016 km², A≈5.64%, B≈99.28%; coarse report ranges 5–7% and 95–100% reflect line/coast uncertainty. The calibrated western corner and two real boundary bends follow the printed outline; markers/annotations were not water exclusions. [Reconstructed polygon](#)

